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A non-linear approximation to the distribution of total expenditure distribution of cruise tourists in Uruguay

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ABSTRACT

This study contributes to the literature on the determinants of tourism spending on cruises at a microeconomic level, through the application of innovative methodologies framed within the machine learning literature. The objective is to study the distribution of the total expenditure of cruise passengers in Uruguay, using data of the 2016–2017 cruise season survey (collected by the Ministry of Tourism of Uruguay). Due to the nature of this variable, we implement a two stages modeling strategy. In the first stage, we model the probability of spending, and in the second, the strictly positive spending. The paper analyze the distribution of conditional expenditure to a set of sociodemographic, travel, contextual and satisfaction variables applying non-linear regression techniques with Lasso penalty and nonparametric techniques such as Random Forest. The empirical results show that the key variables that determine the average spending of cruise tourists are their residence and the port of arrival of the cruise. The analysis of the predictive performance of the models (applied through a training sample and a test sample) shows that Random Forest method has the greater predictive capacity. Finally, the importance variable is analyzed by Random Forest.

1. Introduction

The contribution of the tourism sector to the economies is so important that it manifests itself as the main economic sector in many countries Song, Dwyer, Li, and Cao (2012). In addition to this, it has to be noted the importance of the tourism sector as an engine of economic growth for most of the countries. This topic has been extensively studied in the so called Tourism Led Growth Hypothesis (TLGH), as is shown in recent review papers (see Brida, Cortes-Jimenez, and Pulina (2016); Castro-Nuño, Molina-Toucedo, and Pablo-Romero (2013) and Pablo-Romero and Molina (2013).) This phenomenon is the main motivation for the increasing interest from academicians on studying the economic impacts of tourism. From this economic perspective, one of the main topics that have to be analyzed and studied is tourism demand. The analysis and interpretation of elements such as the demand elasticity or the average expenditure of tourists associated with specific characteristics can provide valuable information for the development of the tourism economy. One strand of this research topic, the analysis of aggregated demand, has been extensively analyzed. There are hundreds of empirical studies using different econometric and statistical methodologies and several published review papers on aggregate tourism demand modeling and forecasting (see Dogru et al. (2017), Peng et al. (2014), Peng et al. (2015), Lin and Song (2015), Dwyer et al. (2012), Divisekera (2013), Song and Li (2008) and Lim (2006)). However, the academic work that analyze the determinants of tourist spending at the individual (microeconomic) level has been relatively less than those studying aggregate (macroeconomic) demand, considering both the number of contributions and the heterogeneity, of statistical methods and econometric models. As highlighted by Wang and Davidson (2010), apart from other published forty years ago (for example see Mak, Moncur, and Yonamine (1977)). Research on this topic starts in the 1990s and has produced no more than 27 studies after 2000.

According to Brida and Scuderi (2013), the interest towards the study of the determinants of individual tourist expenditure is linked to the evolution of the tourist market after the 1990s. Before then tourism was seen as a mass phenomenon, in the fashion of the Fordist economic development framework, when products came with rigidity and high standardization. As pointed in the review papers Mayer and Vogt (2016), Brida and Scuderi (2013) and Xiao and Smith (2006), most of the papers studying individual tourist expenditure have been case studies for specific tourism destinations. Therefore, global conclusions, referring to a larger population cannot be drawn. As it is explained in Brida and Scuderi (2013) different measurements of expenditure, models, dependent variables and regressors are employed in these

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studies and then particular results cannot be generalized. This last comprehensive review reports that various individual characteristics have been tested in models by the past studies, such as economic attributes (income, preferences, occupation, etc.), sociodemographic characteristics (age, gender, place of residence, education, marital status, etc.), trip-related features (accommodation type, transport, destination, travel information source, etc.) and psychographic variables (opinions, attitudes, motivations, etc.). In addition, the same paper sustains that many different research questions are still open. In particular, the authors show that the studies on the determinants of individual tourist expenditure are not characterized by great heterogeneity in the use of statistical econometric models.

Statistical and econometric models involving tourist expenditure and a set of covariables can be categorized into two main groups, according to the scale of measurement of the response variable: models for metric response (for example, linear regressions and its generalizations) and models considering categorical responses (like logistic regressions). Very recent publications introduce innovative and original new methodologies to study the determinants of tourist spending. Some examples of these alternative methodologies are described below. Marrocu, Paci, and Zara (2015) introduces the quantile regression approach to analyze the main determinants of tourist expenditure of non-resident tourists who spent their holidays in Sardinia during the period April–October 2012. Olya and Mehran (2017) uses complexity theory and fuzzy set qualitative comparative analysis to explore a variety of configurations sufficient for simulation of both high and low scores of outbound tourism expenditures. As an alternative to conventional OLS and Tobit estimators, Melstrom (2017) presents an exponential model of tourist expenditures estimated by a quasi-maximum likelihood (QML) technique. Abbruzzo, Brida, and Scuderi (2014) introduces the use of scad-elastic net in the assessment of the determinants of individual tourist spending.

This paper contributes to the strand of the empirical literature on the determinants of individual tourist expenditure. The main research question deals with the study of the impact of a set of socioeconomic and behavioural variables on individual spending, applying an appropriate model in order to overcome multicollinearity and perform selection of regressors. Our analysis runs on the behavior of cruise tourists in Uruguay during the cruise season 2016–2017, by using data from the official survey conducted in Uruguay by Ministry of Tourism (Ministerio de Turismo (2017)). To our best knowledge, this is the first attempt in the microeconomic tourism literature to use this technique. As shown by the comprehensive review of Brida and Scuderi (2013), previous papers apply classic econometric models. Cruise tourism has experienced a very important expansion over the past twenty years, with an average annual growth rate of around 8% in the number of worldwide cruise passengers during the period 1995–2010 (see Paoli et al. (2017); Wang, Wang, Zhen, and Qu (2016); Paoli et al. (2017) and Wang et al. (2016)). In Uruguay, the number of cruise tourists has passed from around 56 thousand (2004–2005) to more than 400 thousand tourists-cruisers in 2011–2012 and 2012–2013. (see Brida, Bukstein, and Tealde (2015); Brida et al. (2015)). The privileged position of Montevideo and Punta del Este ports, located between Buenos Aires and Rio de Janeiro, is a determinant variable in the number of cruises that have as stopping point such ports, even if it should be mentioned that in the last seasons visitors declined but continued to outperform the 300 thousand.

The paper is organized as follows. Section 2 describes the database and the main variables generated from the survey questionnaire. section 3 presents the methodology and section 4 shows the empirical results and discusses them. Finally, section 5 concludes. The codes in the R language (TeamRCore (2017)) are available upon request to the authors.

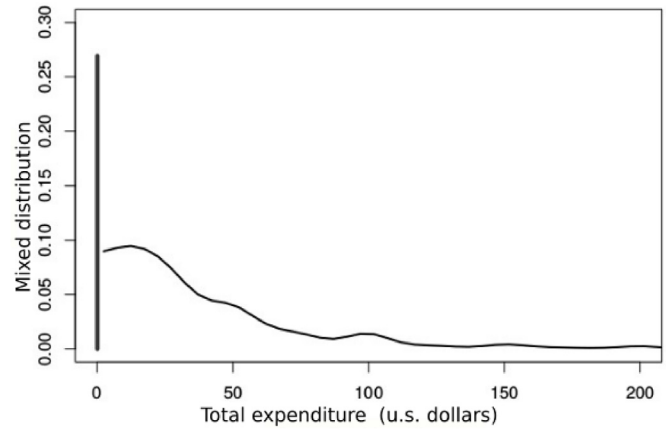

Fig. 1. Representation of the probability function of the mixed variable Total Expenditure. Source: own elaboration with the R language.

2. Dataset description

Individual data came from the 2016–2017 season survey of cruisers, collected by the Ministry of Tourism of Uruguay. The arrivals are distributed between November 2016 and April 2017. Data is open and available at Ministerio de Turismo (2017). The sample was obtained through a systematic sampling on the expected arrivals of the cruisers; in each of them, a simple random sample of tourists is taken. The number of registered surveys was 3117, out of a total of 22 cruise ship arrivals (in Montevideo and Punta del Este ports). The questionnaire includes sociodemographic, context and satisfaction inquiries.

The model is constituted by a nonnegative dependent variable *Total per capita expenditure*. This is a mixed type variable, which distribution is partially discrete and partially continuous, with a positive probability at zero, see Fig. 1.

Mean value of tourist expenditure in American dollars is $ 31, and its standard deviation is $ 35. Nevertheless, excluding null expenditure −27% of the passengers has zero expenditure-the mean value became $ 42.38 with a standard deviation of $ 36.

The explanatory variables can be classified in four different groups: satisfaction variables (variables of pleasure and displeasure), socio–-economic variables (residence, age, occupation), travel variables (month, number of previous visits, descending port, sites visited) and context variables (number of group members -one, two and more than two- and group type -mixed, only women or only men-). The satisfaction variables are dichotomous variables, where the tourist determines which were his main pleasures and dislikes during her/his visit at the destination. A summary of those 10 variables with the highest response rate is presented in Fig. 2. Note that tourists appreciate the gentle treatment (54% of agree), and have a low displeasure for the level of prices of the country (12%).

Fig. 3 shows the distribution of the total expenditure of visitors according to the expense of the cruisers of the sample discriminated according to their nationality and port of landing. Note that Brazilian tourists are those who present a higher level of expenditure, particularly in Montevideo. Also note that a significant proportion of passengers from Argentina and Brazil tend to spend less when the port of landing is Punta del Este. This behavior could be a consequence of the availability of free activities at that destination (beautiful beaches) and the high level of prices. (according to Fig. 2, this is the main source of displeasure mentioned by cruise passengers).

In Fig. 4 it can observed the evolution of Total average expenditure longitudinally, i.e. as a function of the months of the season. Note that lowest average expenditure in the port of Montevideo (less than 25 dollars) was in December 2016, which was also a low-spending month for Punta del Este. Anyway, in the latter, the worst performance was in

Pleasures	Percentage_Pleasures	Displeasures	Percentage_Displeasures
People	54.03	Nothing	72.54
Beach	46.17	Prices	12.09
Building	44.18	No Stores	3.02
Hygiene	41.16	Hygiene	2.34
Tranquility	38.88	Exchange Rate	1.51
Landscape	38.69	Attention	1.28
Port market	37.44	Deterioration	0.74
Montevideo	34.94	No Bathrooms	0.58
Old town	34.49	Poverty	0.55
Food	34.42	SignsTransit	0.51

Fig. 2. Variables that present greater percentage of answer in reference to the Pleasures and Displeasures. Source: own elaboration with the R language.

March 2017. The average of expenditure in November in Punta del Este is null because no cruise ship arrives in this month.

Fig. 5 shows that in general those who spend more are who travel alone, and that the association between average expenditure and age changes according to the type of travel group. If the group integrates with more than two people, then the expenditure grows with the average age of the members of the group. If the group is integrated with a couple, the expenditure would reach a maximum when the average age of the group is between 40 and 50 years. Who travel alone show the peculiarity that those who spend less are just those between the ages of 40 and 50, showing a higher expenditure who are older than 60 years.

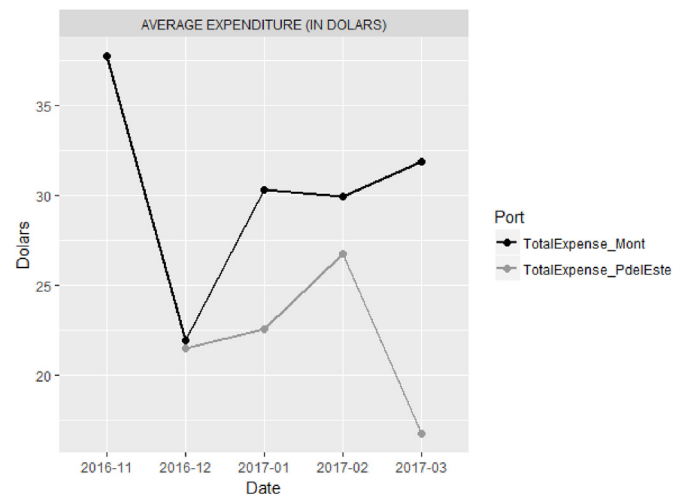

Fig. 4. Average Total Spending in dollars compared to the months of the season according to the port of disembarkation. Source: own elaboration with the R language.

3. Methodology

There are many possible approaches to the analysis of data that contain a large number of zeros. The appropriate methodology depends on the generating source of zeros in the data. Following Humphreys (2013), it is important to understand the difference between of the concepts of truncation data and censoring data. These concepts can represent two different frames for the treatment of zeros in the data. The proposed estimators change according to the adopted approach. Censoring data is when an observation is incomplete due to some random cause. The cause of the censoring must be independent of the

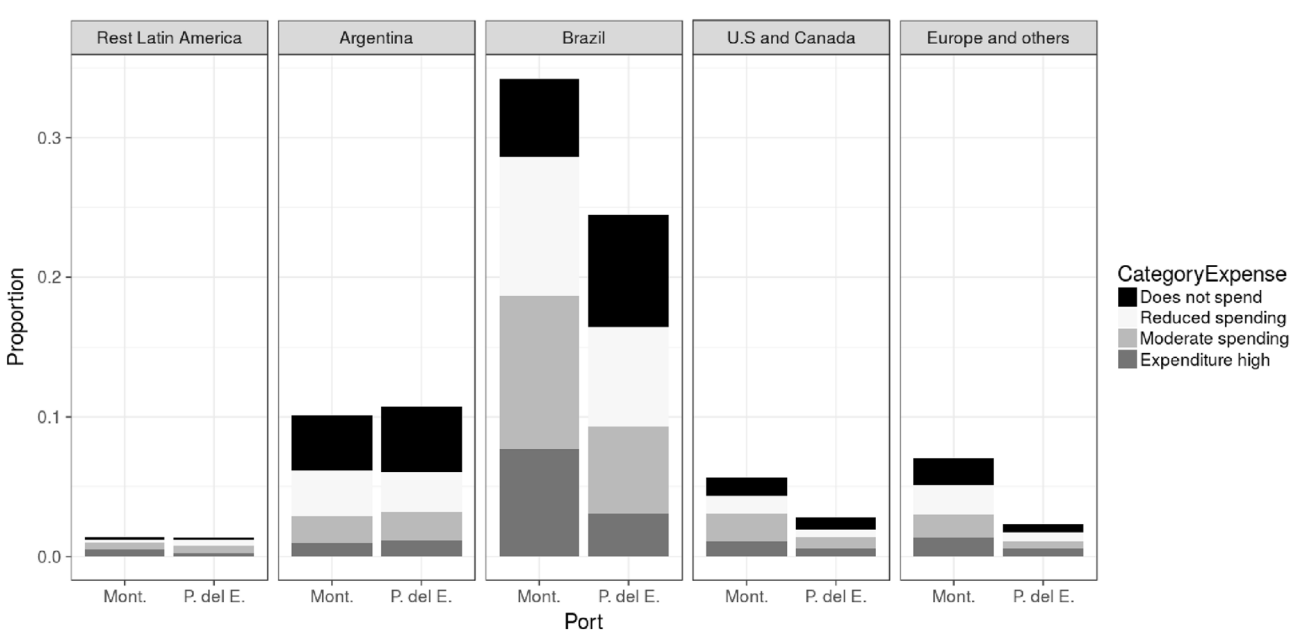

Fig. 3. Proportion of tourists by nationality and in what proportion they spend according to their residence and port of disembarkation. Source: own elaboration with the R language.

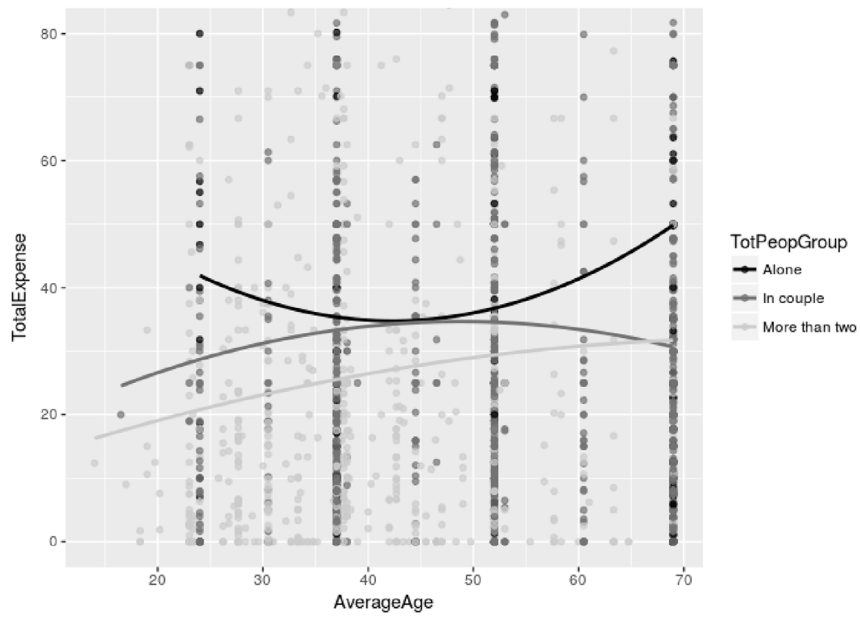

Fig. 5. Association between expenditure and age changes according to the type of travel group. Source: own elaboration with the R language.

event of interest if we are to use standard methods of analysis. Truncation is a variant of censoring which occurs when the incomplete nature of the observation is due to a systematic selection process inherent to the study design. If the observation y_i contain a large number of zeros we can assume that they are generated for two reasons: $y_i = 0$ because the variable was not observed (an appropriate reference for this approach is the Heckman selectivity models, see Heckman (1979)) or, $y_i = 0$ because the variable is a genuine zero. On the latter case two models are used, Hurdle Models and Two-Part Models, see Cragg (1971) and Aitchison (1955) respectively. Following the reasoning of two-stage models, two methodologies will be applied in each stage: regression model with Lasso (Least Absolute Shrinkage and Selection Operator) penalty (Lasso) and Random Forest (RF). For each step, we have a large set of explanatory variables of socio-demographic character, satisfaction and context. When many explanatory variables are available, it is necessary to select variables to obtain a parsimonious model with high predictive power. The subset selection methods (best subset, forward stepwise and backward stepwise selection) involve using least squares to fit a linear model that contains a subset of the predictors James, Witten, Hastie, and Tibshirani (2013). Our innovation consists in framing, each of the stages, in the machine-learning paradigm. That is, we work with an alternative methodology consisting of in each step, fit a model containing all p predictors using a technique that constrains or regularizes the coefficient estimates, or equivalently, that shrinks the coefficient estimates towards zero. Shrinking the coefficient estimates can significantly reduce their variance. The two best-known techniques for shrinking the regression coefficients towards zero are Ridge regression and the Lasso. Motivated by the objective of finding a linear regression technique that was stable but that made selection of variables, in Tibshirani (1996) is proposed Lasso. This is one of the penalty technique for the coefficients of the variables, which has the advantage of allowing regression and selection at the same time. The coefficients of a Lasso model, are estimated by maximum likelihood adding a penalty term to the optimization problem, that is equivalent to,

$$\min_{\boldsymbol{\beta}} L(\boldsymbol{\beta}|\boldsymbol{x}) + \lambda \sum_{j=1}^{p} |\beta_j|$$

where $\boldsymbol{\beta} = (\beta_1, ..., \beta_p)$, $\boldsymbol{x}$ are the covariables and L here is a loss function (usually a quantity that is proportional to a negative log likelihood, such as the residual sum of squares).

The optimization algorithm depends on an implicit parameter, which we denote as λ, considered the penalty factor. As λ grows the amount of variables that the model will take as significant will be greater (less contraction). Through cross-validation, we study the goodness of fit of the model as the λ penalty parameter. It is observed that as this parameter grows, the number of predictor variables decreases. Since the coefficients can be estimated exactly equal to zero it is said that the Lasso models also perform selection of variables.

As a second alternative methodology to apply in each stage, it is proposed Random Forest. This methodology is based on Classification and Regression Trees (CART). The objective of CART consists in the determination of predictions of new observations from the stratification (segmentation) of the space of predictor variables (covariables) according to certain criteria of partition. CART was developed by Breiman in Breiman (1996) and Breiman (2001), a current reference is James et al. (2013). The iterative algorithm takes into account three definitions for its development:

1. A partition criterion dividing the root node (given by the set of all the observations of the sample with the label of the group to which each belongs) in two sub samples (child nodes), so that child nodes are more homogeneous in relation to the group they belong to than the preceding node,
2. A stop criterion, to consider when a terminal node (leaf) is obtained and the process stops and
3. A label (in classification) o quantitative measure (regression) is assignment criterion of the terminal node or leaf.

The main disadvantage of CART is that it is a method poor in robustness, that is, it is sensitive to small changes in the data. To obtain greater robustness can be applied other algorithms such as: Random Forest, Bagging, Boosting. In particular, Random Forest will be applied by combining n trees based on n boostrap samples. The main advantage is given by the improvement in the predicted capacity of the CART, obtaining estimators with less variance, see Breiman (2001). The disadvantage of this methodology is the loss of interpretability, although they have methods to classify the importance of covariables. This technique is complementary to the application of classical regression models, with the advantage of not requiring any a priori assumption (machine-learning method) and its performance is high even when the variables present correlates or complex interactions.

3.1. Application

Let Y be the variable that represents the total expenditure of cruise tourists represented by the sum of the expenses of the following items: food, tours, transportation, purchases and other expenses. This variable is a variable truncated to zero, so we proceed to apply a model in two steps. In the first stage it is modeled the probability of spending and in the second stage the strictly positive total expenditure is modeled. Lasso and RF methodologies are applied in both stages. The results obtained through the before mentioned techniques will be compared with those obtained using a classic regression model (glm), also in two stages with selection of variables. For the implementation of Lasso methodology, proceed as follows: in the first step (Classification Step) a Logit model with Lasso is constructed to obtain a model for the probability that a tourist will spend or not. In second stage (Regression Step), a regression model with Lasso is applied to estimate the average value of the total expenditure *per capita*. In this paper, we consider the largest value of the penalty factor such that classification/prediction error is within one standard error of the minimum. For the RF application, it is based on the first stage in the combination of 500 classification trees (based on 500 boostrap samples) and in the second stage in a similar way combining regression trees.

Finally, both methodologies are combined in the two stages of regression and compared with general linear models (glm). In this last, the selection of variables is performed using the Backward method in the two stages.

4. Results

For the implementation of the models mentioned above, the original sample is partitioned into a training sample and a test sample. The first one is obtained by simple sampling without replacement of 2/3 size of the original sample and the second is obtained as the difference between the first and the original sample.

Models are estimated with the training sample, and with the test sample the prediction error of the models is evaluated, that is, the validation of the model will be carried out from a new set of data, different from those obtained with the model. This procedure was carried out 100 times, taking the training samples and test samples independently in each iteration. Then, in each step, the models are put into competition by comparing the corresponding classification and regression errors.

In each methodology two types of error are computated: in the first step the percentages of misclassified cruise passengers from the confusion matrix, and then a global error is calculated from the application of the two-step models calculated as the root of the error quadratic mean between the expenditure and the expenditure predicted in the test sample, that is,

$$\text{RMSE} = \sqrt{\frac{\sum_{j=1}^{n} (\hat{Y}_j - Y_j)^2}{n}} \quad \text{(Root mean squared error)}$$

here n is the size of the test sample, Y_j is the expenditure of individual j of the test sample and $\hat{Y}_j$ is the estimated value after the two stages of regression. We report the mean error of the 100 iterations. Analogously to the previous step, classification error (in the test sample) of the first stage is analyzed.

4.1. Comparison of predictive classification power in the first stage

In each replication, as we schematize in Fig. 6, the classification errors of the first stage are observed for the test sample (the size is 750).

The glm, Lasso and RF methods are compared. Table 1 shows the confusion matrices for each of the methodologies. The reported values are the average of the total replications of the predicted percentage of false positives and false negatives. We observe that the global prediction error is 28%, 21% and 19% in glm, Lasso and RF respectively. However RF presents a high percentage of false positives, that is, cruise passengers who did not incur expenses and the RF algorithm predicts that they did. Although the prediction errors are high, this is an expected phenomenon because of it is a problem ofthe high complexity to indicate from the measured covariables whether the cruise passenger spends or not.

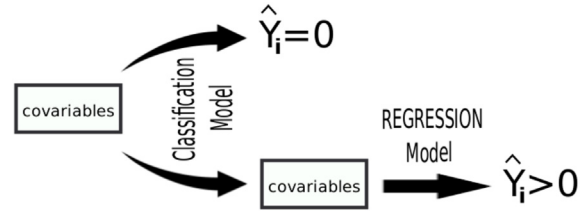

Fig. 6. Representation of the predictions from the two-stage model. Source: own elaboration with GIMP.

Table 1
Classification results in percentage in the test sample (mean in the 100 replications) in the first stage step for the general linear models (GLM), Lasso and RF. Source: own elaboration with the R language.

Prediction	$Y = 0$			$Y = 1$		
	GLM	LASSO	RF	GLM	LASSO	RF
$\hat{Y} = 0$	17%	19%	14%	19%	14%	7%
$\hat{Y} = 1$	9%	7%	12%	55%	60%	67%

4.2. Comparison of predictive classification power in the complete two-stage models

We obtain different predictors that emerges as a combination of the previously applied models. To do this, a global prediction error is calculated from the combination of predictors of the first and second steps. The results are presented in Table 2 in terms of the RMSE (expressed in American dollars).

All cases show relatively high errors with respect to the deviation of the variable Y ($\sigma_Y = 36$). The method that presents the best performance is apply RF in the two stages ($\text{RMSE}_{\text{RF-RF}} = 19.9$). Note that it is not correct to use RF with the original data in a single stage (because the model does not predict spending values equal to zero). In addition, RMSE calculated on bases on RF regression in a single stage ($\text{RMSE}_{\text{RF}} = 23.7$) higher than the RMSE with RF in both stages.

4.3. Variable importance

Based on the previous performance analyses, we computed the importance variable (at each stage independently) by RF. As it was already mentioned in the Methodological section, the Random Forest method is a tree aggregation method, which has the disadvantage of losing interpretability. However, two indicators could be obtained

Table 2
The RMSE (in American dollars) in the test sample (mean in the 100 replications) for the two-stages models combining the different methodologies. Source: own elaboration with R.

		2nd step		
		GLM	Lasso	RF
1st step	GLM	25.2	25.6	24.2
	Lasso	24.7	23.5	23.1
	RF	22.9	21.4	19.9

Table 3
Importance of variables for RF in both stages. We denote MDA to Mean Decrease Accuracy, MDG to Mean Decrease Gini, IMSE the percentage increment of MSE and INP of the increment of node purity. Source: own elaboration with the R language.

First Stage Classification				Second Stage Regression			
Var	MDA	Var	MDG	Var	IMSE	Var	INP
Port	26.4	Residence	39.9	Port	17.2	Residence	143792
Residence	20.7	Port	38.6	Casino (like)	16.5	Month of arrive	143383
Casino (like)	13.1	Occupation	33.7	Residence	14.4	Occupation	116068
Organization(like)	12.5	Month of arrive	32.2	Old City (like)	12.3	Port	84122

regarding the importance of the variables in the successive trees considered for aggregation. Firstly, an average decrease in the mean quadratic error (MDA) of predictions is obtained when a variable is excluded from the model. Additionally, we calculate an indicator of the decrease in the total impurity of trees based on dividing according to that variable (Gini index). Table 3 shows both indexes. Based on this table, the arrival port, residence, occupation, the liking for the casinos and the month of arrival are determining variables in the two stages of the model through RF.

It should also be noted that the core of variables that RF gives greater importance, in most cases, are not discarded either in the Lasso penalty procedures or the backward glm algorithm.

5. Conclusions and future research

In this paper, we applied two alternative methodologies to model the total expenditure of cruise passengers in Uruguay in season 2016–2017 with respect to a set of covariates (sociodemographic, behavioural and satisfaction variables). Applying these different methods allows as analysing the robustness of the results and comparing them even with a classic regression model.

The methodologies are innovative, in the sense that they are techniques framed within Machine Learning. In particular, CART and RF are non-classical models that are usually used in problems where $p \gg n$ and where no a priori assumption is needed. The two-step model presents an innovation when using Lasso to perform regression and model selection at the same time.

The first conclusion that arise from the analysis of competition in predictive performance of the models (applied through a training sample and a test sample) is that Random Forest evidences a greater predictive capacity with respect to the other two models applied.

The results indicate that the RF performance reflects the existence of non-linear associations between the covariables used and the total expense of the cruise tourist. The RF is the most robust methodology when it comes to predicting total tourist spending.

Secondly, results from the methodologies coincide in highlighting the predominant role of residence and port of landing, as the variables more associated with of total expenditure. The first sociodemographic variable may be linked with the purchasing power of the different origins of cruise passengers in Uruguay. The second one (a behavioural variable), may be linked with the opportunities of doing activities that no implies extra expenditures, like resting in the beach when they landed in the port of Punta del Este.

Thirdly, results from two of the methods (Two-Stage Regression with Lasso and Random Forest) also highlight the variables Occupation, Port, Residence and Month of visit as variables that explain the level of total expenditure of tourists.

Results founded in this paper are similar to those found in Bellani (2016) by applying a very different empiric approach: an econometric Heckam selection model. Bellani also highlighted that residence, the fact of not traveling alone, the port of landing and satisfaction are the main variables to explain total expenditure of cruisers in Uruguay (in seasons 2010 to 2014).

A direct line of research that follows to this work is to analyze and compare several cruise seasons using the methodology that yields better results regarding predictive performance. Also in the same schedule of the empirical approach, it may be interesting to analyze the expenditures desegregated in different items: travel expenditure, on shopping, tours, food, etc.

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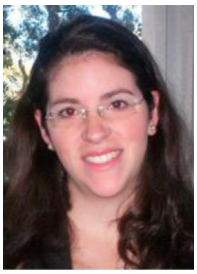

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